

Equality Chains (EQUAL: block)

Example 1 : Base case - length of the empty sequence

(a)

Prove $\text{length } \langle \rangle = 0$.

$$\begin{aligned} & \text{length}(\langle \rangle) \\ &= 0 \quad \text{[by definition of length]} \end{aligned}$$

Example 2 : Inductive step - length of a non - empty sequence

(a)

Prove $\text{length } (s \hat{\ } \langle x \rangle) = \text{length } s + 1$.

$$\begin{aligned} & \text{length}(s \hat{\ } \langle x \rangle) \\ &= \text{length}(s) + 1 \quad \text{[by definition of length and concatenation]} \end{aligned}$$

Example 3 : Multi - step chain with justifications

$$\begin{aligned} & \text{length}(s \hat{\ } t) \\ &= \text{length}(s) + \text{length}(t) \quad \text{[by length - concatenation lemma]} \\ &= \text{length}(t) + \text{length}(s) \quad \text{[commutativity of +]} \end{aligned}$$